

Data Centers: Noise & Air Quality Concerns | Loudoun County, VA - Official Website

Category: Acoustics (dBA vs dBC) & Health Impacts | Evidence ID: #215

Archived: 2026-10-04 20:20

- **Original URL:** <https://www.loudoun.gov/6405/Noise-Air-Quality-Concerns>
 - **Research Category:** Acoustics (dBA vs dBC) & Health Impacts
 - **Archival File:** 215_www_loudoun_gov_6405_Noise-Air-Quality-Concerns.pdf
 - **Extraction Status:** Full Web Transcript PDF
-

Source Content

Data Centers: Noise & Air Quality Concerns Loudoun County has urban, suburban and rural areas, with approximately two-thirds of the county's population living in one-third of the county's land area, in the suburban and urban policy areas in the eastern portion of the county. This area is also home to nearly 200 operational data centers. The mingling of residential and data center development raises the profile of a number of community impacts, including noise concerns and air quality issues. Noise from Data Centers Data center campuses can include a variety of noise sources that generate sounds at a range of frequencies. These include the basic machinery that keeps a data center up and running, including lighting, cooling, mechanical systems and the electricity needed to run them. Noise also emanates from HVAC units and emergency backup generators. Increasingly, data center campuses may feature onsite energy generation and energy storage systems, which also produce noise. Some of the noise generated by data centers and associated uses occurs at mid-range frequencies and is plainly audible to the human ear at normal volumes. Much of this noise is characterized by a persistent hum or buzzing sound and occurs within a narrow range of sound frequencies. Broadly known as tonal noise, single frequency tones can be more aggravating to the human ear than broadband background noise at similar volumes. Data centers and their associated infrastructure can also generate persistent noise near the lower limit of sound frequencies that can be heard by the human ear. Such low frequency noise is less audible at normal volumes, but some studies suggest low frequency sound may pose adverse impacts ranging from general discomfort to potential health complications. The scientific community acknowledges a need for more research to assess the long-term impacts of low-frequency noise. Challenges of Regulating Low-Frequency Noise Regulating noise can be difficult because the production of noise and how it is perceived depends on several factors, including, the time of day, the type of noise being produced, topography, environmental and weather conditions, existence of vegetation or structures, background noise during monitoring activities, expectations of peace and quiet, the context or location of the person hearing the noise, whether water is nearby and more. Tonal noise and low frequency noise pose specific measurement and regulation challenges. Loudoun County and many other localities have historically used decibel-based A-weighted sound measurement standards to regulate noise (dBA) because dBA measurements focus on the noise frequencies most audible to humans; however, dBA measurements generally do not capture low frequency noise or the discomfort caused by steady tonal sounds. The typical maximum sound levels employed in local noise regulations measuring by dBA are often too high to account for pervasive tonal noise. The most effective approaches to measuring and regulating such noise use multiple measurement types, such as octave band and C-Weighted decibel measures (dBC),

which some localities utilize to better monitor and control noise across a range of frequencies. Regulations based on dBA measurements are generally easier to implement and enforce, while regulating noise by additional sound level measurements require specialized training for that type of measurement and equipment that can measure those sound levels. In Loudoun County, both the county's codified ordinances and a portion of the zoning ordinance govern noise from industrial or commercial sources, though the zoning ordinance is more often used to address noise violations from business operations. Per the Loudoun County Zoning Ordinance, data centers are required to submit a pre-construction noise study at the time of site plan review as well as a post-construction noise study submitted at least one month but no more than 12 months after the issuance of the first Certificate of Occupancy. The Loudoun County Zoning Ordinance also stipulates maximum sound levels, measured in dBA at the receiving property line, for different zoning districts. For residential properties, the maximum allowable sound level at the property line is 55 dBA. When a resident submits a complaint, the county's zoning enforcement division responds and collects data to evaluate the noise emanating from the suspect noise source. If a property is found to be generating noise at a level in violation of the ordinance, the offending party is required to demonstrate compliance with regulatory mandates. Historically, data centers have been cooperative and collaborative with addressing noise concerns. However, should a substantiated violation lack curative responses from the offending party and remain unabated, zoning enforcement staff initiate progressive enforcement action through the issuance of a Notice of Violation and subsequent civil penalty fines, as appropriate. Noise Complaints Noise complaints, which residents may submit online, are among the most persistent issues brought forward to Loudoun County officials and frequently pertain to tonal noise. The county does not currently have policy or regulatory language specific to tonal noise or low frequency noise. As part of the Data Center Standards and Locations project initiated by the Board of Supervisors in 2024, the county is working to identify new policy and regulatory mechanisms to address data center noise, including tonal noise and low frequency noise. Amendments to the county's Comprehensive Plan and Zoning Ordinance are anticipated in spring or summer 2027. Changes in State Law for Future Applications A recently adopted statute, codified as Code of Virginia § 15.2-2209.5, requires legislative land use applications to include a site assessment and sound profile when an applicant reasonably anticipates that proposed facilities will require 100 megawatts or more of electrical power from an electric utility. The county is then expressly required to assess consistency with the Comprehensive Plan, Zoning Ordinance and other applicable laws and regulations. Air Quality Concerns Air quality is of concern to county residents, especially when data centers in the county run on backup power for extended periods of time. The majority of data centers in Loudoun County run their backup diesel generators for one hour, once per week, for mandatory testing and may turn to backup power on the rare occasion there is an interruption to their power supply from the grid or if ordered to do so by the regional transmission organization, PJM, due to extreme electrical demand on the grid. There is only one data center in Loudoun that is reliant on off-grid power full time, and that facility runs on natural gas turbines. Diesel generators produce air pollutants such as nitrogen oxides, particulate matter, carbon monoxide and sulfur dioxide. With appropriate after-treatment filters, generators can achieve near-zero emissions, but not all of Loudoun County's data centers employ these filters. A majority of data centers in Loudoun County use Tier 2 generators, some of which are retrofitted with filters to improve air quality. Tier 4 generators are far less common and are equipped with filtration systems to drastically reduce pollution. Loudoun County does not have the authority to regulate exterior air quality conditions; that falls under the purview of the Virginia Department of Environmental Quality (DEQ). On July 1, 2026, DEQ began mandating Tier 4-equivalent emissions standards for new or modified generators at data center sites. The new guidance from DEQ regulates nitrogen oxides, carbon monoxide, and particulate matter. More information is available from the revised DEQ guidance memo on the topic. DEQ is also currently working to evaluate air quality in northern Virginia near "data center alley" in Ashburn with the objective

of determining if ongoing air quality monitoring is necessary in areas with high concentrations of data centers. More information on the project, including near-real-time sensor readings, is available online.